MILITARY SPECIFICATION

SLEEVE AND GRAIN ASSEMBLY (CKU-2/A Rocket Catapult)

This specification is approved for use by NAVSEASYSCOM, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of Sleeve and Grain Assembly referred to herein as the grain assembly.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-0082662A	Catapult, Aircraft Ejection Seat, CKU-2/A and CKU-5/A Assembly
MIL-L-82653	Liner Materials, Polybutadiene Base
MIL-P-82680	Propellant Solid Composite (CKU-7/A and CKU-2/A Rocket Catapults)

STANDARDS

FEDERAL

FED-STD-313	Material Safety Data Sheets, Preparation and the Submission of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Division (501), Indian Head, MD 20640, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1377

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MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-453	Inspection, Radiographic
MIL-STD-810	Environmental Test Methods

DRAWINGS

Naval Sea Systems Command (Code Ident 53711)

5184311	Shipping Container For Rocket Catapults
5184312	Shipping Container, Inner Packing For Rocket Catapults

Naval Ordnance Station, Indian Head, MD (Code Ident 14083)

10551586	Sleeve and Grain Assembly
11743866	Catapult, Aircraft Ejection Seat CKU-2A Assembly
11743884	Sleeve (Grain)

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Code of Federal Regulations

49 CFR 100-199	Transportation

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402. Orders for the above publication should request the latest edition and supplements thereto.)

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3. REQUIREMENTS

3.1 First article. When specified in the contract (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.4).

3.2 Conformance to documents. Materials, parts and assemblies shall conform to the applicable drawings and referenced specifications.

3.3 Formulation and materials. The propellant formulation shall be in accordance with MIL-P-82680, Class II. The liner formulation shall be inaccordance with MIL-L-82653 Type I.

3.4 Environmental. Each grain assembly when assembled with all other components of the CKU-2/A catapult shall meet the environmental requirements of MIL-C-0082662A with no evidence of disintegration of propellant or derangement of components. In addition each assembly shall meet the ballistic requirements of MIL-C-0082662A when assembled into rocket catapult hardware conforming to drawing 11743866.

3.5 Cured propellant physical properties. The physical properties of the cured propellant shall be in accordance with MIL-P-82680 Class II.

3.6 Material safety data sheets. The contractor shall prepare and submit material safety data sheets in accordance with FED-STD-313 as specified in the contract.

3.7 Workmanship. The workmanship shall be of sufficiently high quality to ensure satisfactory operation, reliability, and durability consistent with the requirements and application of the charge assembly and in accordance with the following:

3.7.1 Voids. Total void volume shall not exceed 0.250 cubic inch per grain. Propellant void maximum dimensions shall not exceed 0.375 inch. Interconnected voids shall be considered as a single void.

3.7.2 Cracks/Unbonds. There shall be no cracks in the propellant and no unbonds between the propellant and the liner or the liner and the sleeve. The sleeve shall be free of separation, voids, blisters, fissures, burrs, sharp corners and unfilled areas.

3.7.3 Foreign matter. The assembly shall not contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements

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specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5)

4.3 Test conditions. Unless otherwise specified, all tests required by this specification shall be performed in an atmospheric pressure of 28 to 32 inches of mercury, a temperature of 60 to 90°F, and a relative humidity of not more than 90 percent.

4.4 First article inspection.

4.4.1 Sample. Unless otherwise specified in the contract (see 6.2), a first article sample of 18 grain assemblies is required and shall be submitted to an activity designated by the procuring activity (see 6.2) for inspection. The grains shall be assembled with the required inert components to form the rocket catapult CKU-2/A, drawing 11743866. Sample grain assemblies shall be manufactured using the methods and procedures proposed for production. Any production of the grain assemblies by the contractor, prior to approval of the sample, shall be at the contractor's risk. First article samples shall not be included as a part of the quantity specified for delivery in the contract (see 6.2).

4.4.2 Inspection. The first article inspection is conducted for the purpose of determining prior to starting production, that the contractor's production methods are capable of producing grain assemblies that comply with the technical requirements of the contract. The first article inspection shall consist of the physical properties test of 4.8.1 and the first article tests specified in MIL-C-0082662A. Failure of any grain assembly to comply with any requirement of MIL-C-0082662A or this specification shall be cause for rejection of the first article sample.

4.4.2.1 Sample failure. If a failure, a malfunction, an out-of-tolerance condition, or an inability to obtain results due to loss of data occurs during or after a first article test, the supplier will determine analytically the most probable cause of non-compliance. If the non-compliance is unique to the unit tested, the test may be repeated (in full or in part) with a substitute unit and testing continued on the remainder of the program. If this analytical examination reveals the suppliers production techniques to be inadequate (see 4.4.2), the production inadequacy must be corrected and all or part of the first article sample testing as defined in 4.4.2 repeated as explained in 4.4.2.2 at the discretion of the Government.

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4.4.2.2 Retest. At the discretion of the procuring activity, first article sample tests, or any portion thereof, shall be repeated under the following conditions:

a. The manufacturer has modified his product through a change of raw materials, production procedures, or methods. The supplier shall notify the procuring activity prior to the incorporation of any such changes, and shall provide quantitative evidence of the effect of such changes on the performance or characteristics of the product. The requirement for retesting will be based on an evaluation, by the procuring activity, of the evidence supplied.

b. There is evidence that the quality of the product has not been maintained. This evidence may be in the form of accumulated failure reports of the product, of system failures attributable to the product, or failure of the product to pass any of the tests for quality verification that may be conducted by or for the procuring agency.

c. Any change in design or documentation by the procuring activity.

d. Any interruption of production by the contractor.

4.5 Quality conformance inspection. The quality conformance inspection shall consist of the examination of 4.7 and the tests of 4.8 performed on samples selected per 4.6.

4.6 Sampling for quality conformance inspection. Unless otherwise specified, sampling shall be in accordance with MIL-STD-105.

4.6.1 Lot size. For the purpose of identification and shipment, a lot shall consist of a maximum of 1200 grain assemblies, exclusive of static firing test samples, provided the homogeneity of the inspection lots has been retained as follows: ingredients from the same lot of raw materials. A production lot shall be that number of grain assemblies manufactured with no change in design, construction or manufacturing techniques from those used in fabricating the first article sample, and which are to be presented for acceptance at one time. Each production lot shall contain assemblies cast from a single propellant mix.

4.6.2 Sampling for examination. Sampling for examination for major and minor defects shall be in accordance with TABLE I.

4.6.3 Sampling for tests.

4.6.3.1 Sampling for physical properties test. Samples for the cured propellant physical properties tests of 4.8.1 shall be prepared for each lot in accordance with MIL-P-82680.

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4.6.3.2 Sampling for functional tests. Twenty-one sample grain assemblies shall be selected from each lot for testing in accordance with 4.8.2. These samples shall be forwarded to the activity designated by the procuring activity for test (see 6.2).

4.6.3.3 Sampling for radiographic examination. Each grain assembly shall be radiographically inspected in accordance with 4.8.3.

4.7 Quality conformance examination. Examination for major and minor defects shall be performed on a defect basis using the sampling plan and associated criteria in TABLE I in accordance with the classification of defects contained in drawings 10551586 and 11743884. If the total number of defects per classification for either the major or minor classification exceeds the acceptance number of TABLE I, the lot shall be rejected. All nonconforming material or parts shall be removed from the lot. Four power (4X) magnification may be used in performing visual examination for such defects as burrs or foreign matter.

TABLE I. Sampling plan.

Lot size	Sample size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination	-	-	-	-
301 - 800	50	0	1	1	2
801 - 1200	90	1	2	2	3

4.8 Quality conformance tests.

4.8.1 Cured propellant physical properties. The cured propellant physical properties shall be determined in accordance with MIL-P-82680 Class II. Compliance with the requirements of MIL-P-82680, Class II shall be demonstrated for the first article sample and each individual lot during production. Failure to meet the requirements shall be cause for rejection of the lot.

4.8.2 Functional test. Sample units selected according to 4.6.3.2 will be static fired by the Naval Ordnance Station, Indian Head in accordance with MIL-C-0082662A. Failure of any unit to meet the ballistic requirements of MIL-C-0082662A will result in rejection of the lot.

4.8.3 Radiographic examination. Each grain assembly shall be radiographically inspected, while at ambient temperature, in two mutually perpendicular longitudinal planes in accordance with MIL-STD-453 to verify compliance with 3.7.1 and 3.7.2. Each grain assembly shall be inspected at ambient temperature after being maintained at that temperature for 4 hours.

4.9 Measuring and test equipment. The supplier shall provide for the selection, evaluation, approval, maintenance and control of all inspection standards, gages, measuring and test equipment necessary to determine conformance to this specification.

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4.10 Instrumentation accuracy. Unless otherwise specified, the test instrumentation systems shall be such as to yield data of the following accuracies. The percentage figures represent the limits of error expressed in percent of expected maximum value.

a.	Temperature	±2°F
b.	Barometric Pressure	±5 percent
c.	Pressure	±5 percent
d.	Time	±5 percent for functional tests

5. PACKAGING

5.1 Preservation, packaging and packing. Packing shall be level A in accordance with Drawings 5184311 and 5184312.

5.2 Marking. In addition to any special markings required by the contract (see 6.2), marking shall be in accordance with MIL-STD-129, MIL-STD-130, and 49 CFR 171-179.

6. NOTES

6.1 Intended use. The sleeve and grain assembly described herein is intended for use in conjunction with CKU-2/A Aircraft Ejection Seat Catapult Assembly in accordance with MIL-C-0082662A.

6.2 Ordering Data. Procurement documents should specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this specification
b. Quantity required exclusive of first article and static fire sample units (see 3.1, 4.4 and 4.6.3.2)
c. When a first article sample is not required (see 3.1, 4.4 and 6.4)
d. Assigned activity for first article testing (see 4.4) and quality conformance inspection functional testing (see 4.6.3.2 and 4.8.2)
e. Any special markings required (see 5.2)
f. Safety precautions (see 6.3.2)

6.2.2 Contract data requirements. Any data required for delivery in connection with this document shall be specified on a DD Form 1423 incorporated into the contract. Such data will be delivered as identified on completed (numbered) DIDs (Data Item Descriptions/DD 1664) which will be documented in the applicable ADL (Authorized Data List).

6.3 Explosive hazard classification. The explosive hazard classification for this grain assembly shall be:

Explosive Hazard Classification: Class 2
Storage Compatibility Group: Group J
Department of Transportation (DOT) Hazard Class: Class B
DOT Marking: Propellant Explosive

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6.3.1 Hazard notice. The rocket described herein is flammable and/or explosive and consequently presents a hazard in manufacture, handling storage and shipment. The contractor should recognize this hazard and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling and electrically induced incidents.

6.3.2 Safety precautions. The safety precaution requirements of the "Contractor's Safety Manual for Ammunition, Explosives, and Related Dangerous Material" (DOD 4145.26M) are applicable and should be specified in the contract or order as required by the Defense Acquisition Regulations (DAR) 1-323.

NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore" (OP 5) should be made applicable.

6.4 First article sample waiver. First article samples submitted and approved on a recent contract may be accepted by the procuring activity in lieu of an additional first article inspection. When the first article sample is waived (see 6.2.1(c)), the procurement document should contain a statement specifying that the standards of workmanship exhibited by the previously approved first article sample shall determine the minimum requirements of the current contract.

6.5 Disposition of hardware, scrap propellant, and metal parts. The contract should specify that all items purchased become the property of the U.S. Government; no hardware, propellant, or metal parts acceptable, scrapped, or tested shall be disposed of without authorization of the procuring activity; all items be made available to the procuring activity upon request; and that the disposal of any tested hardware, propellant, or metal parts without authorization by the procuring activity will result in the test being declared invalid, and a retest will be required.

6.6 Definitions.

6.6.1 Propellant batch. A propellant batch is a mix composed entirely of one homogeneous lot of ingredients.

6.6.2 Propellant lot. A propellant lot is one or more propellant batches all having the same ingredient lots.

6.7 Suggested source of supply. TP-H-9001; Propellant, Solid, Composite and TLH-317A; Liner, as manufactured by Thiokol Corporation, Longhorn Division, Longhorn Army Ammunition Plant, P. O. Box 1149, Marshall, TX 75670, are products that have met the requirements of this specification in past procurement actions. This information is for the convenience of the procuring activity and is not to be construed as a waiver of any requirement of this specification nor as any limitation of additional potential sources of supply.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-S-82714(OS), Sleeve and Grain Assembly (CKU-2/A Rocket Catapult)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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